

# The Illusion of Optimality: Challenges of increasing complexity in multi-risk assessment

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## Abstract

Efforts to integrate multiple hazards, exposure and vulnerabilities into coherent multi-risk frameworks have advanced over the past three decades, yet the field continues to lack operationalisable metrics to assess whether increasing modelling complexity has improved predictive or decision-making efficiency. This paper argues that the bias–variance tradeoff, originally from statistical learning theory, can serve as an analytical lens to interpret this trajectory and to ask whether the field’s growing complexity reflects genuine progress or an “illusion of optimality”. Without explicit measures of informational efficiency, risk integration could become performative rather than functional. We acknowledge that the path toward optimality must be reconciled with the actionability required by practitioners, and that once social components become central to an assessment, what counts as “optimal” is no longer technically determined but becomes relational, context-dependent, and contested. The paper therefore does not prescribe a universal optimum. Instead, it defines informational efficiency as the marginal gain in decision-relevant output obtained for each added increment of model complexity, and proposes a multi-dimensional set of markers—statistical, decisional, temporal, and institutional—through which this gain can be assessed.

## 1 Introduction

Multi-risk assessments are inherently complex, requiring the integration of compounding hazards, dynamic exposure, and evolving societal vulnerabilities. Over the past three decades, tools and methods have advanced significantly to address these multiple layers of complexity. The bias–variance framework (Geman et al., 1992) provides a pragmatic analytical argument

for a long-standing challenge in risk science: determining when integration enhances understanding and when it merely amplifies uncertainty. Rooted in statistics and popularised in machine learning, the concept describes the tradeoff between underfitting (high bias) and overfitting (high variance) (Geman et al., 1992; Hastie et al., 2009). Lower predictive error implies reduced uncertainty over outcomes, which in turn affects the expected decision (Domingos, 2000). Building on Dunant (2021), who first applied this framework as an analytical argument toward multi-hazard risk assessments, this paper extends the analysis in three directions: (i) a three-phase historical mapping of the field’s trajectory onto the bias–variance continuum, (ii) the introduction of *informational efficiency* as a core evaluative concept, and (iii) a diagnosis of the “illusion of optimality”—the risk that increasing complexity is mistaken for progress in the absence of shared efficiency metrics.

We specifically refer throughout to “multi-risk” frameworks as defined by Zschau (2017): integrated systems that account for complex interactions among multiple hazards, exposed assets, and societal vulnerabilities—as distinct from simpler multi-hazard approaches that consider co-occurrence without systemic interdependencies. It is important to specify that regardless of the terminological scope—whether strictly multi-hazard (physical processes), multi-hazard risk (asset exposure), or multi-risk (systemic societal vulnerability)—the fundamental structural challenge remains analogous with the understanding that the rate of complexity increases at each step. Indeed each step up this hierarchy introduces new parameters, thereby increasing the risk of variance (overfitting) if the underlying process understanding does not scale at the same rate as the model’s complexity.

As Dunant (2021) highlighted, single-hazard models in multi-hazard environments exhibit high bias, while overly complex multi-risk models might show high variance without proportional gains in predictive accuracy (Figure 1). However, this framework has remained largely theoretical. The field lacks operationalisable metrics—quantitative where measurable, qualitative where not—to situate itself within this landscape. The central premise of this paper follows directly: *we do not know where the field currently sits on the bias–variance curve*. It may be close to an optimal balance, still on the high-bias side, or already drifting toward inefficiency; none of these positions can presently be confirmed or ruled out.

This paper argues that *informational efficiency* should become a core evaluative dimension of multi-risk science. We define it as the *marginal* gain in uncertainty-aware, decision-relevant output obtained for each added increment of model complexity—for example, one additional hazard layer, interaction, or stakeholder input—relative to the simpler model without it. Framing efficiency at the margin is deliberate. The relevant question is not whether a complex model is useful overall, but whether each addition earns its place, which is the logic of value-of-information analysis (Howard, 1966; Keisler et al., 2013) and of the distinction between forecast quality and forecast value (Murphy, 1993). The concept further

draws on decision theory (Horvitz and Barry, 1995), information theory (Shannon, 1948; Cover and Thomas, 2006), and risk communication research (Klinke and Renn, 2002). In the context herein, informational efficiency is distinct from traditional statistical accuracy. In multi-risk environments, a purely quantitative “performance” metric risks missing the local social features and contexts that are fundamental to risk decision-making. The marginal gain can therefore be expressed in several currencies—statistical, decisional, temporal, and institutional—rather than a single rigid metric; candidate quantifications are proposed in Table 1. We acknowledge that explicitly measuring efficiency may not in itself dissolve the illusion of optimality, particularly when social systems resist the very notion of a clear-cut and universal optimum. The argument is therefore not that better metrics will solve the problem, but that without any shared language of efficiency, the field cannot even diagnose where it stands.

## 2 Conceptual Framework

The bias–variance framework is used here, as in Dunant (2021), as a conceptual heuristic—not as a literal statistical mapping. In statistical learning theory, bias and variance are mathematically decomposable components of prediction error; in multi-risk science, they are not. The analogy is therefore diagnostic rather than quantitative: it provides a vocabulary for asking whether a given integration effort evolves toward oversimplification (high bias) or toward unmanageable complexity (high variance).

The concern is not new to environmental modelling. The difficulty of validating open-system models against an unambiguous ground truth (Oreskes et al., 1994), the equifinality of competing model structures (Beven, 2006), and the risk of getting “the right answers for the wrong reasons” (Kirchner, 2006) have long motivated calls for parsimony and iterative model evaluation (Box, 1976; Jakeman et al., 2006). More recently, Puy et al. (2022) showed across a large sample of environmental and epidemiological models that models with higher effective dimensions tend to produce more uncertain estimates, and Saltelli et al. (2020) warned that complexity pursued for its own sake can undermine the relevance of models to the societies they serve. The opposite position has also been argued: over-simplified models can hide structural error and underestimate predictive uncertainty (Hunt et al., 2007). In machine learning, the classical U-shaped risk curve has itself been qualified by the “double descent” behaviour of heavily over-parameterised models (Belkin et al., 2019). We note this caveat explicitly, because it cautions against assuming that more complexity *must* mean more variance. In multi-risk assessment, however, the data regimes that make double descent possible are rarely available, and the relevant costs of complexity are as much cognitive and institutional as they are statistical. Behavioural research makes a parallel point: simple

heuristics can outperform complex strategies precisely because they trade a little bias for a large reduction in variance (Gigerenzer and Brighton, 2009; Gigerenzer and Gaissmaier, 2011).

This lens applies beyond physical models. In participative exercises, the pursuit of comprehensive social integration may introduce redundancy, conflicting inputs, or epistemic noise. While the accumulation of heterogeneous and sometimes incommensurable knowledge claims from diverse disciplinary and stakeholder perspectives is a valuable reflection of complex social realities (Schlüter et al., 2024; Rusca et al., 2026), from a purely operational standpoint, it can challenge an actionable synthesis—thereby diminishing informational efficiency without commensurate gains in immediate decision relevance. This complexity also directly undermines interpretability: if practitioners cannot trace the causal logic amidst the noise of competing narratives, they cannot assume the professional responsibility required to translate assessments into action (Pereira Santos et al., 2024).

In this context, the pursuit of “multi-risk” optimality must account for bounded rationality (Simon, 1956, 1947) and acknowledge the absence of “unlimited resources of time, information, and computational might” (Gigerenzer, 2020). Analogous to the “efficiency–resilience” tradeoff (Hynes et al., 2020), hyper-integration (low bias) could produce an “integration–resilience tradeoff”, where noise overtakes the capacity for action.

The following sections categorise the evolution of multi-risk science into three chronological phases and read them against the bias–variance continuum (Dunant, 2021) (Figure 1). An important caveat applies. The phases describe the *direction* in which the field has moved—toward more integration and more parameters—not its *position* on the curve. For this reason, Figure 1 deliberately places the phases beside the error curve rather than on it: without measures of informational efficiency, we cannot say whether any phase lies to the left of the optimum, at it, or beyond it.

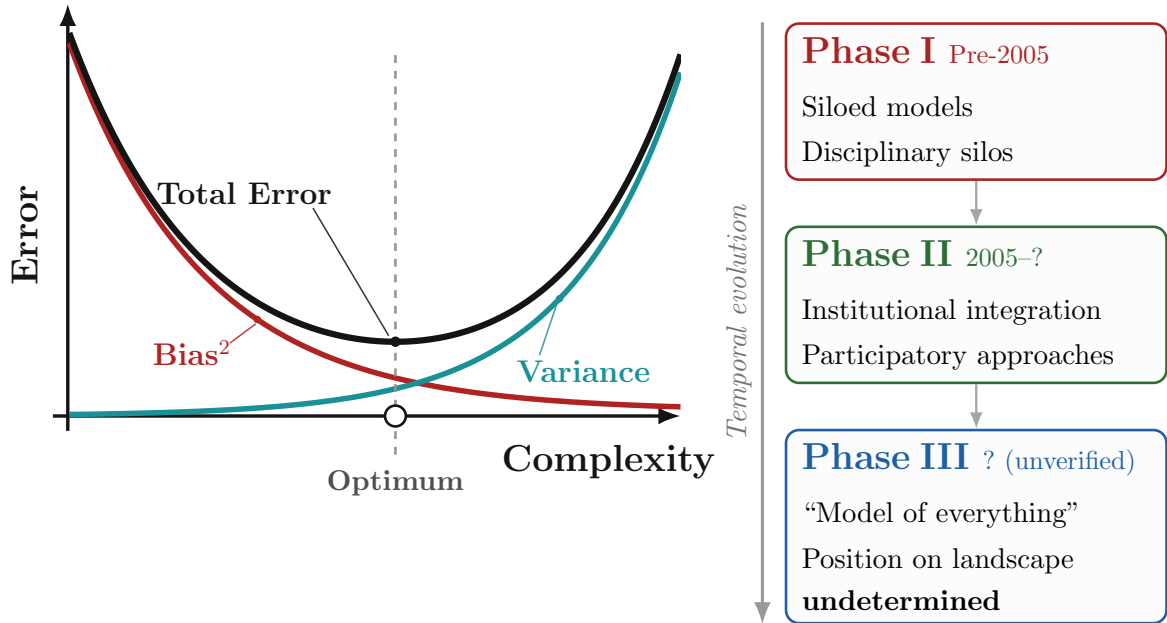


Figure 1: Conceptual framing of the bias–variance landscape for multi-risk assessments.

This framing illustrates the direction of travel from the high-bias simplicity of siloed origins toward the potentially high-variance complexity of modern hyper-integration. It does not locate the field on the curve; instead, it frames the question of whether current integration efforts achieve functional efficiency or merely drift toward an “illusion of optimality”.

### 3 Evolution of multi-risk science

#### 3.1 Phase I: Siloed Origins and Emerging Integration Mandates (before 2005)

The first attempts to formalise multi-hazard assessment emerged alongside early international frameworks promoting integrated disaster management. The idea itself is older: Hewitt and Burton (1971) had already argued for studying “all hazards at a place” rather than one hazard at a time. On the science side, the volcanology community provided significant early precedents, as volcanic crises inherently involve the interaction of multiple simultaneous hazards like ashfall, lahars, and seismic activity (Tilling, 1989; Blong, 1996); event trees were developed to structure these interacting outcomes during crises (Newhall and Hoblitt, 2001). These were, however, exceptions driven by the physics of a single hazard source rather than a general practice across the field.

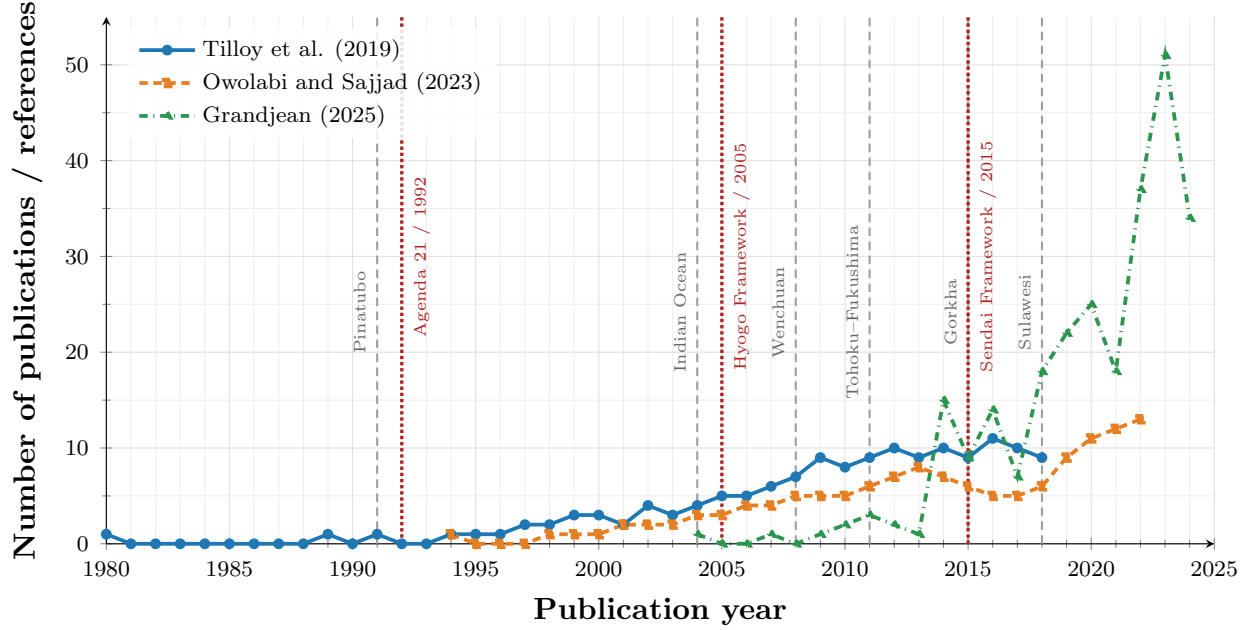


Figure 2: Timeline synthesis of annual publication/reference counts for multi-hazard and multi-risk research. Tilloy et al. counts were rebuilt by aggregating the **Year** column from the supplementary workbook; Owolabi and Sajjad counts were rebuilt by aggregating the **Year** column from the supplementary material; Grandjean is kept as a contextual 2004–2025 annual distribution from Figure 3 in Grandjean (2025), who frames the selected corpus over 2004–2025. Grey dashed lines mark selected major multi-hazard or cascading events since 1990: Pinatubo eruption–lahars (1991), Indian Ocean earthquake–tsunami (2004), Wenchuan earthquake–landslides (2008), Tohoku earthquake–tsunami–Fukushima (2011), Gorkha earthquake–landslides (2015), and Sulawesi earthquake–tsunami–liquefaction (2018). Red dotted lines mark Agenda 21 (1992), the Hyogo Framework (2005) and the Sendai Framework (2015). Publication counts indicate research activity and institutional momentum; they are not a measure of model complexity or performance.

The principles of sustainable development outlined in *Agenda 21* (UNEP, 1992) and later reinforced by the Johannesburg Plan (Nations, 2002) and the Hyogo Framework for Action (UN-ISDR, 2005) established the groundwork for multi-hazard thinking. Figure 2 shows the institutional momentum that followed: annual publication counts from three independent bibliometric reviews (Tilloy et al., 2019; Owolabi and Sajjad, 2023; Grandjean, 2025) rise slowly before the Hyogo Framework and accelerate after it, and again after Sendai. These counts track research activity, not model complexity or performance; they show how quickly the field grew, not whether that growth moved it toward the optimum.

During this phase, risk assessments remained largely disciplinary, emphasising individual

hazards such as floods, earthquakes, or landslides. Studies like Middelmann and Granger (2000) and Granger et al. (1999) in Australia exemplified early local-scale multi-hazard assessments, but these typically overlaid independent single-hazard analyses rather than modelling interactions between them (Kappes et al., 2012; Gill and Malamud, 2014). In bias–variance terms, most models of this period sat on the high-bias side: simplified assumptions and a focus on single-event probabilities, regardless of whether events occurred in a multi-hazard environment.

Uncertainty was rarely quantified explicitly, and hazards were often combined through qualitative or semi-quantitative matrices (Kappes et al., 2012; Gallina et al., 2016). The resulting models were interpretable but detached from cascading dynamics and long-term systemic risks.

### **3.2 Phase II: Institutional Integration and the Illusion of Optimality (2005–?)**

Following the Hyogo Framework, institutional momentum accelerated the integration of multiple hazards. The Sendai Framework (UNDRR, 2015) later reaffirmed this mandate, emphasizing "all-of-society" and "systemic" approaches. These international mandates, by prioritizing 'holistic' and 'all-hazard' frameworks, effectively incentivized the creation of comprehensive systems designed to map every interaction. Conceptual work by Marzocchi et al. (2009, 2012) and Kappes et al. (2012) articulated the need for frameworks addressing multi-hazard and multi-risk interactions, since single-hazard approaches in multi-hazard environments are inherently biased toward specific features and miss critical interdependencies. Systematic classifications of hazard interactions (Gill and Malamud, 2014, 2016) and reviews of multi-risk methodologies (Gallina et al., 2016) gave this agenda a common vocabulary. The transition from single-hazard to systemic multi-risk assessments is, in part, driven by what Zschau (2017) and Hochrainer-Stigler et al. (2024) identify as the twin challenges of risk globalisation and climate change. Globalisation has rendered risks interdependent through cross-linked critical infrastructure and global supply chains, while climate change alters the frequency, intensity and spatial distribution of extreme events (Seneviratne et al., 2021), including compound events whose joint occurrence drives impacts (Zscheischler et al., 2018, 2020) and consecutive disasters that strike before recovery is complete (de Ruiter et al., 2020). These drivers create "high-risk dynamics" that require capturing various interactions (Pescaroli and Alexander, 2018), and in doing so intensify the bias–variance dilemma highlighted herein.

Research in the modelling space during this phase sought to reduce bias by linking hazard chains, formalising interactions statistically (Selva, 2013; Mignan et al., 2014), introducing

Bayesian networks, and incorporating network-based or spatio-temporal frameworks (Tilloy et al., 2019; De Angeli et al., 2022; Terzi et al., 2019; Dunant et al., 2021, 2025). These efforts represented a major step toward a unified understanding of cascading effects and feedback loops. In bias–variance terms, they were designed to reduce bias; whether they also increased variance, and by how much, was not measured.

Yet the apparent convergence toward optimality might be deceptive. Several reviews show that integration often increased conceptual overlap but still struggled to translate complexity into actionable indicators (Owolabi and Sajjad, 2023; Laino et al., 2024; Hochrainer-Stigler et al., 2024; Scolobig et al., 2017). Feedback from civil protection stakeholders on multi-risk decision-support tools pointed to the same gap between what the tools offered and what practitioners could use (Komendantova et al., 2014), a pattern also found in broader reviews of decision support systems for natural hazards (Newman et al., 2017). The field seems to achieve visibility but not necessarily efficiency—that is, the marginal gain in decision relevance or process understanding obtained for each added increment of complexity. The illusion of optimality persisted not simply because standardised metrics were absent, but because the very notion of what constitutes an informational gain was never made explicit: whether efficiency should be measured in terms of improved risk communication, facilitated action, or deeper process understanding remained unarticulated. This ambiguity—not merely a technical gap—is what could allow increasing complexity to be mistaken for progress. Dunant (2021)’s warning that as model complexity increases, the risk of “overfitting” escalates, potentially impairing the practical precision of multi-hazard outputs, is even more pertinent today as the tools for large-scale computation and collaboration become more accessible (Mack, 2011).

### **3.3 Phase III: Rising Complexity and the Unquantified Present (?)**

While the recent proliferation of highly complex, socially integrated models might suggest the field has already transitioned into a “Phase III” of multi-risk assessment, we argue that this boundary remains entirely unverified. Because the field lacks operationalised metrics for informational efficiency, we simply cannot know whether the current landscape represents a genuine paradigm shift toward optimal risk integration, or merely an inflation of variance and epistemic noise. Consequently, Phase III remains an unquantified horizon rather than a confirmed reality. Nevertheless, we argue that the field (i.e. multi-hazard, multi-risk) stands at a crossroads today. Advances in computational modelling, remote sensing, and machine learning have enabled “hyper-integrated” frameworks spanning climate, geophysical, and social domains (Ferrario et al., 2025; MYRIAD-EU, 2022; Lundstrom and Alam,



2022). Conceptual frameworks for complex and systemic risk now explicitly call for interactions between hazards, between responses, and across sectors and scales to be represented (Simpson et al., 2021; Renn et al., 2020; Hochrainer-Stigler et al., 2023; Ward et al., 2022), while vulnerability itself is increasingly treated as dynamic rather than fixed (de Ruiter and van Loon, 2022). However, these systems risk overfitting: absorbing massive input data into interconnected systems without tangible, measurable improvement in output quality or decision usability.

As demonstrated by recent assessments of European transboundary risk management, the primary hurdle to operationalising multi-risk science is not a lack of model complexity, but a lack of fundamental harmonisation in metrics and data structures across borders (Polese et al., 2024). Workshops with European stakeholders likewise identified governance, the translation of science to policy and practice, and data availability—rather than modelling capacity—among the main barriers to multi-hazard risk management (Šakić Trogrlić et al., 2024). While scientific “super-models” attempt to simulate complex hazard interactions, practitioners are often still restricted by a “fragmented data landscape” and a lack of common vulnerability classes, which makes the outputs of highly integrated models difficult to compare or legally validate in an operational context (Polese et al., 2024). Furthermore, as Kalogiannidis (2024) highlight, the effectiveness of disaster management in Europe is more significantly hindered by administrative and legislative obstacles—such as the lack of standardised protocols—than by a lack of computational integration. This suggests that the “illusion of optimality” in complex modelling fails to account for the “social reality” of the legal and administrative constraints within which Civil Protection agencies must operate.

Informational efficiency, in the marginal sense defined above, remains largely untracked: few studies report what an added hazard layer, interaction, or stakeholder input contributes relative to the simpler model without it. Without such evidence, integration can drift into epistemic opacity, where uncertainty and redundancy coexist. As Dunant et al. (2021) emphasised, finding a “healthy medium between accuracy (bias) and precision (variance)” requires quantifying the relationship between model complexity, risk and action.

Future progress depends on being honest about what kind of progress is being made. The objectives of *scientific progress* (advancing process understanding) and *operational progress* (supporting defensible decisions) are not opposed, but they optimise differently and require different standards of evaluation. Conflating them risks producing the “Mushy Middle” (Stonehouse and Snowden, 2007): work that achieves neither genuine process insight nor novel operational application—sophisticated enough to appear rigorous, tractable enough to appear useful, but evaluated against neither standard. For the researcher, the relevant question is therefore not whether a model is complex, but whether its complexity resolves a specific gap in process understanding that simpler models cannot. For the practitioner,

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the question is whether it narrows the range of defensible decisions under uncertainty in a way that is institutionally actionable (Polese et al., 2024). In both cases, the threshold for “good enough” is as much social and institutional as it is technical—it depends on who will use the output, under what constraints, and toward what decision (Hanger-Kopp et al., 2022; Simon, 1956). Research on usable science reaches the same conclusion: information is taken up when it is credible, salient, and legitimate to its users (Cash et al., 2003), and when producers and users interact to close the “usability gap” (Dilling and Lemos, 2011; Lemos et al., 2012; Kirchhoff et al., 2013). Where the supply of scientific information is not matched to the demands of decision-makers, more research does not translate into better decisions (Sarewitz and Pielke, 2007; Weichselgartner and Kasperson, 2010).

What the field currently lacks might not be more integration, but an explicit, shared account of these thresholds. Without it, efficiency claims remain unverifiable, complexity continues to be equated with progress, and the illusion of optimality persists (Scolobig et al., 2017; Kalogiannidis, 2024).

## 4 Discussion

The three phases outlined above trace a field moving from high-bias simplicity toward greater complexity, without a clear mechanism for evaluating its actual position on the curve or whether this trajectory constitutes progress. Several observations emerge.

First, the bias–variance analogy illuminates but does not resolve. In statistical learning, model selection can be guided by cross-validation against held-out data; in multi-risk science, an equivalent ground truth against which to calibrate is rarely available, because the systems being modelled are open and the events of interest are rare (Oreskes et al., 1994). The analogy therefore functions as a diagnostic vocabulary—useful for framing the problem, but insufficient as a prescriptive tool.

Second, the distinction between scientific and operational progress—raised in Phase III—deserves further attention. The “Mushy Middle” (Stonehouse and Snowden, 2007), originally a concept from competitive strategy describing firms that pursue neither cost leadership nor differentiation, finds a parallel in multi-risk initiatives that achieve neither deep process understanding nor operational tractability. Recognising this risk does not require abandoning integration, but it does require that researchers and practitioners articulate—before model development—which standard of evaluation applies.

Third, while informational efficiency is defined here as a marginal quantity, its *measurement* is deliberately left open. Premature formalisation risks privileging one definition of “useful output” over others. Nevertheless, the field would benefit from pilot studies that attempt to quantify efficiency in bounded contexts—for example, by comparing the marginal

decision value of adding a second hazard layer to a single-hazard assessment in a specific governance setting. Well-established tools already exist for parts of this task, from value-of-information analysis (Howard, 1966; Keisler et al., 2013) to the economic value of forecasts under cost–loss decisions (Richardson, 2000; Pappenberger et al., 2015). Table 1 summarises candidate metrics across four dimensions for such efforts.

Table 1: Dimensions of informational efficiency for multi-risk assessments. Rather than a single scalar metric, efficiency is operationalised as a multi-dimensional evaluation spanning statistical, decisional, temporal, and institutional domains.

| Efficiency Dimension        | Core Decision Question                                                                      | Candidate Metric / Proxy                                                                                                                                                                                                                                                                                                                                                                                                                                          | Theoretical Origin                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Information Gain</b>     | Does adding this complexity (e.g., a hazard layer) reduce uncertainty about the risk state? | <i>Mutual Information</i> : Measures whether a new model input (e.g., an additional hazard layer) genuinely reduces uncertainty about the risk outcome, or merely adds noise.<br><i>Output Uncertainty Change</i> : Change in the spread of model outputs once that input is included, relative to a simpler baseline; added inputs may narrow or widen it.                                                                                                       | (Shannon, 1948)<br>(Varley, 2025)<br>(Puy et al., 2022)            |
| <b>Decision Impact</b>      | Does the added complexity actually change the decision that would be made?                  | <i>Decision Divergence Rate</i> : Frequency with which the complex model triggers a different categorical action (e.g., evacuate vs. shelter) compared to a simpler heuristic.<br><i>Expected Value of Including Uncertainty (EVIU)</i> : Change in expected loss when acting on the complex model vs. a baseline.<br><i>Relative Economic Value</i> : Value of the model to a user with a given cost–loss ratio, relative to climatology and a perfect forecast. | (Morgan and Henrion, 1990)<br>(Murphy, 1993)<br>(Richardson, 2000) |
| <b>Temporal Efficiency</b>  | Does the model yield the decision within the available operational time window?             | <i>Decision Latency</i> : Time elapsed from data input to an actionable, interpretable output.<br><i>Time-to-Decision Window</i> : Ratio of model processing time to the physical lead time of the hazard.                                                                                                                                                                                                                                                        | (Reichstein et al., 2025)                                          |
| <b>Institutional Uptake</b> | Can the end-user actually process and defend the output under pressure?                     | <i>Utilisation Rate</i> : Proportion of offered model runs that are actually used in the final decision (a compliance measure).<br><i>Cognitive Load</i> : Number of distinct probabilistic inputs a decision-maker must synthesise to reach a binary                                                                                                                                                                                                             | (Simon, 1956)<br>(Miller, 1956)<br>(Hart and Staveland, 1988)      |

None of these markers is sufficient on its own, and not all will apply in every setting. Testing them in bounded contexts—where the decision, its users, and its time window are known—is the obvious next step.

## 5 Conclusion

The history of multi-risk assessment shows a clear direction of travel—from siloed, high-bias assessments toward increasingly integrated and complex ones—but not a known position. Whether the field today sits short of the bias–variance optimum, near it, or beyond it cannot be established with the evidence currently collected. This perspective suggests that increasing model complexity does not necessarily improve understanding or decision relevance, as integration can introduce additional uncertainty and interpretability challenges (Puy et al., 2022; Saltelli et al., 2020). Recent advances in holistic frameworks and AI-driven systems highlight both the potential and the difficulty of managing these trade-offs (Ward et al., 2026; Reichstein et al., 2025).

Building on the bias–variance perspective introduced by Dunant (2021), this paper argues that the central challenge is not only methodological but epistemic. Greater integration must be accompanied by clearer research intent and a systematic evaluation of informational efficiency—namely, the marginal gain in uncertainty-aware, usable output that each added increment of complexity delivers. The bias–variance framework therefore serves not as an end, but as a diagnostic tool to assess whether added complexity yields meaningful gains.

To address this, we propose a multi-dimensional set of informational-efficiency markers (Table 1). Each compares a model with a simpler version of itself, and together they span statistical, decisional, temporal, and institutional gains. They do not define a universal optimum, but they give the field a shared way of asking whether an added layer of complexity has earned its place.

Establishing such approaches is essential to ensure that the evolution of multi-risk modelling remains both rigorous and actionable in the context of increasingly interconnected global risks.

## Acknowledgements

## Competing Interests

The authors declare no competing interests.

## Data Availability

The annual publication counts shown in Figure 2 were compiled from the supplementary material of Tilloy et al. (2019) and Owolabi and Sajjad (2023) and digitised from Figure 3 of Grandjean (2025). The compiled counts are provided in the Supplementary Material.

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